

END SEMESTER ASSESSMENTS, SPRING 2024

Program: B.TECH COMPUTER SCIENCE & ENGINEERING / B.TECH COMPUTER SCIENCE & BUSINESS SYSTEMS / B.TECH COMPUTER SCIENCE & ENGINEERING (LATERAL) / B.TECH COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING) / B.TECH COMPUTER SCIENCE AND ENGINEERING (IOT, CYBERSECURITY AND BLOCKCHAIN TECHNOLOGY)		
Subject Name: DATABASE MANAGEMENT SYSTEM		
Subject Code: 1000013103 / 1000023103 / 1000033103 / 1000103103 / 1000113103		Semester: IV
Duration: 2 Hours		Maximum Marks: 50

Note: Answer the following questions as directed.

Group A

(Answer All questions)

(Each question carries 1 mark)

1. Write the correct answer from the options given:

(10 X 1)

(i) Identify which of the following is not a type of DBMS model?

- ☒ (a) Hierarchical (b) Relational
(c) Object-Oriented (d) Sequential

(ii) Mention the purpose of a foreign key in a relational database?

- (a) It uniquely identifies each row in a table (b) It establishes a link between two tables
☒ (c) It enforces referential integrity (d) It defines the primary key for a table

(iii) The projection operation in relational algebra is used for

- (a) Selecting rows based on a condition ☒ (b) Selecting columns from a relation
(c) Combining two relations (d) Removing duplicate tuples from a relation

(iv) In which normal form are composite attributes and multi-valued dependencies removed?

- (a) First Normal Form (1NF) (b) Second Normal Form (2NF)
(c) Third Normal Form (3NF) ☒ (d) Fourth Normal Form (4NF)

- (v) Relational algebra is
- ☒ (a) A procedural query language
 - (b) A non-procedural query language
 - (c) A scripting language
 - (d) A programming language
- (vi) Mention ACID property ensures that either all operations of a transaction are completed successfully or none of them are applied?
- ☒ (a) Atomicity
 - (b) Consistency
 - (c) Isolation
 - (d) Durability
- (vii) Identify the following operations that can typically be performed on views?
- (a) Insertion
 - (b) Deletion
 - (c) Update
 - ☒ (d) All of the above
- (viii) State the purpose of a trigger?
- ☒ (a) To enforce data integrity
 - (b) To improve database performance
 - (c) To control access to the database
 - (d) To create database backups
- (ix) Identify the concurrency control technique ensures that transactions are executed in a serial order?
- (a) Two-phase locking
 - (b) Timestamp ordering
 - (c) Optimistic concurrency control
 - ☒ (d) Serializability
- (x) Define which of the following is not a type of database index?
- (a) B-tree index
 - (b) Hash index
 - (c) Binary index
 - ☒ (d) Bitmap index

Group B

(Answer any Four questions out of Six)
(Each question carries 5 marks)

(4 X 5)

2. Explain the three-schema architecture of DBMS.
3. Define different types of anomalies which occur during referential integrity. Explain it with proper example. (3 + 2)
4. Explain briefly about object-oriented data model.

5. Define BCNF. Why it is considered stronger than 3NF? (2 + 3)

6. Consider the Employee database:

Emp(Ename, Street, Eid)

Works(Ename, Com_name, Salary)

Company(Com_name, City)

Managers(Ename, Mname)

Write the appropriate SQL statement on the basis of above table

(a) Find the name of employee who has got 2nd highest salary.

(b) Find all employees in the database who don't work for SBI.

(3 + 2)

7. Normalize the following schema with given constraints to 4NF:

Books(b_no, isbn, title, author, publisher)

Users(user_id, name, dept_id, dept_name)

b_no → isbn

isbn → publisher, author, title

user_id → author, name, dept_id

dept_id → dept_name.

Group C

(Answer any Two questions out of Four)

(Each question carries 10 marks)

(2 X 10)

8. Outline different database users? Discuss five main advantages of database management system over file management system. (5 + 5)

9. Given a relational schema Supply (sno, city, status, pno, qty) with FD set $F = \{sno \rightarrow city, city \rightarrow status, \{sno, pno\} \rightarrow qty\}$ Find the candidate key of the schema. Also reduce it into 3NF. (4 + 6)

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10. Analyse the deadlock and starvation problem in database in concurrent transaction? Explain the ACID property of Transaction in detail. (6 + 4)

11. Write a short note (any two) (5 + 5)

(a) Lossless-join and Lossy join dependency

(b) Serializability

(c) Data Independence.